# Project R.O.C.K.

Resilient Optimization of Clinker Kinematics

## PROBLEM STATEMENT

The cement industry faces a massive logistical puzzle: efficiently moving raw clinker from manufacturing plants (Integrated Units) to processing facilities (Grinding Units). Currently, planners rely on rigid, deterministic models that assume perfect market conditions. However, real-world demand is highly volatile. When unexpected market fluctuations occur, this static guesswork breaks down, forcing companies to rely on costly emergency shipments, face stockouts, or hoard excessive safety stock.

Furthermore, the environmental toll of these inefficiencies is staggering. Clinker production is incredibly energy-intensive, and wasteful logistics such as running half-empty emergency trucks or overproducing safety stock that risks spoilage severely inflates the industry's carbon footprint. Ultimately, the inability to dynamically adapt to uncertainty forces companies into a lose-lose situation: bleeding capital on inefficient logistics while simultaneously failing to meet critical corporate sustainability and carbon-reduction targets.

## PROJECT OVERVIEW

We are developing a two-tier, AI-driven planning system that expects the unexpected. Ultimately, we are transforming clinker allocation from a manual, error-prone process into a dynamic, resilient supply chain strategy that optimizes resources, absorbs market shocks, and drives verifiable reductions in carbon pollution.

## SOLUTION OFFERED

The proposed solution offers an integrated, AI-driven architecture designed to proactively manage supply chain volatility. First, it deploys an advanced machine learning forecasting model (utilizing XGBoost) to predict baseline clinker demand at consumption nodes while rigorously capturing market fluctuations.

Rather than relying on a single, rigid forecast, our multi-period stochastic optimization engine leverages these predictive insights to compute the optimal routing and production schedules across thousands of probabilistic demand scenarios. By automatically factoring in complex operational parameters—namely, Minimum Shipment Batch Quantities (SBQ) and strict integer constraints for transport routing—the system mathematically minimizes a dual-objective function: end-to-end logistics costs and transportation-related $CO_2$ emissions.

## END USERS

**Primary Users:** Supply chain planners and logistics managers who will use the automated engine daily to schedule cost-effective, constraint-compliant clinker shipments.

**Secondary Users:** Plant operators and sustainability officers who rely on the model's predictive outputs to optimize inventory, prevent stockouts, and track carbon emission reductions.

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## TECHNOLOGY USED

Our solution leverages a robust **Python**-based technology stack to seamlessly integrate machine learning with mathematical optimization. Data processing and feature engineering are handled using **Pandas** and **NumPy**, which feed into our predictive AI layer powered by **XGBoost** and **Scikit-learn** to generate probabilistic demand scenarios. The core Operations Research engine is built with **PuLP** (utilizing solvers like CBC or Gurobi) to process these stochastic scenarios and solve the complex mixed-integer routing constraints. Finally, the entire pipeline is supported by a **PostgreSQL** database and deployed through an interactive **Streamlit** web dashboard, utilizing **Plotly** to visualize optimal supply chain routes, carbon emission tracking, and real-time cost reductions.